

Wireless Physical Layer, Channel, and Capacity

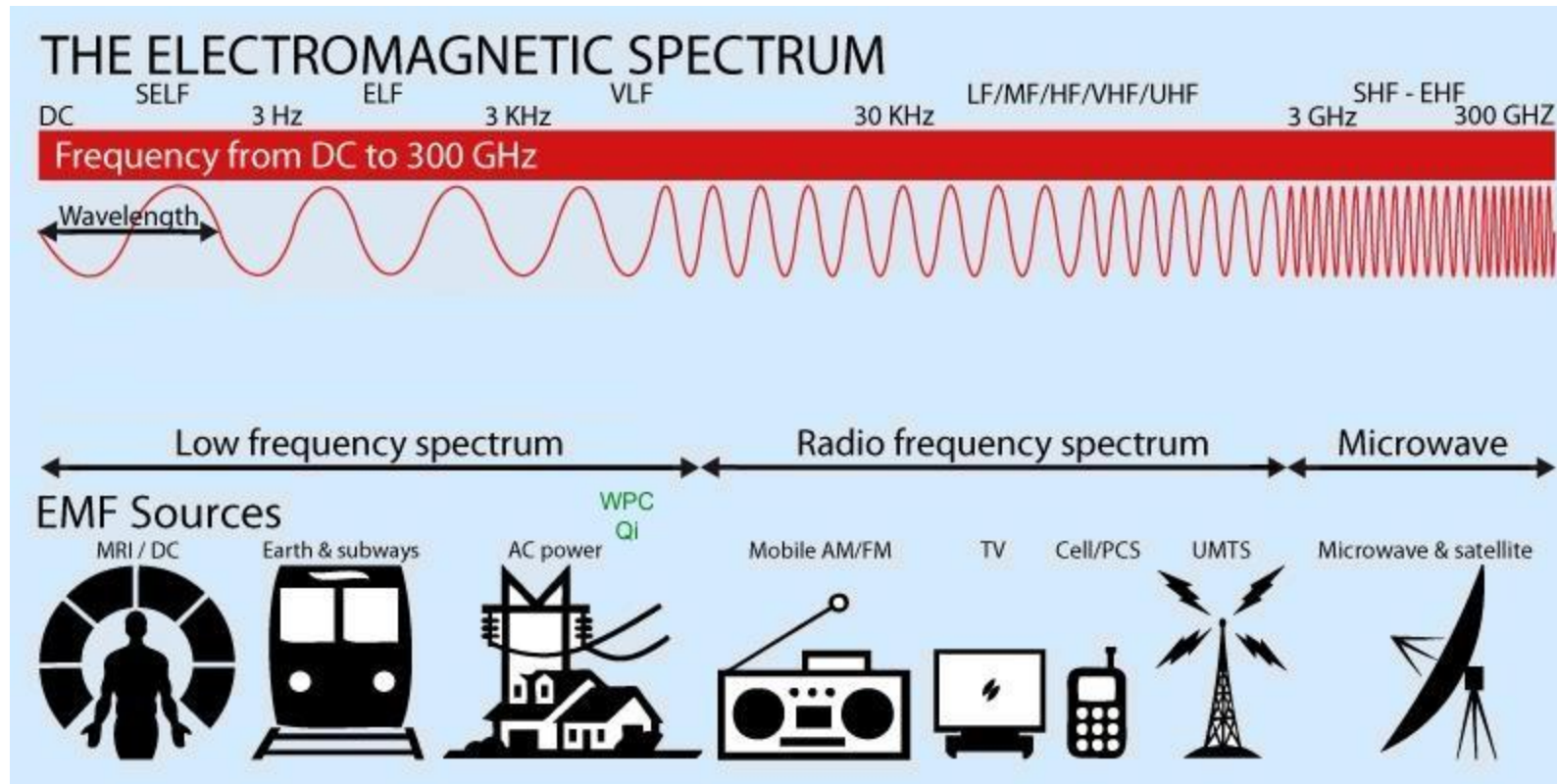
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Radio waves and spectrum

- Physical layer of mobile systems – wireless using electromagnetic (EM) waves
- EM spectrum

EM spectrum



Radio waves and spectrum

- EM spectrum

Name	LF (Low frequency) MF(Medium) HF(High)	VHF (Very High)	UHF (Ultra High)	SHF (Super
Frequency	30Hz-30MHz	30-300 MHz	300Mhz-3GHz	3-30 GHZ
Wave length (speed of light/frequency)	Few Km to metres	Few metres	Few hundreds of cm	Few cm
Used by	Radio, submarines	Analog TV	Digital TV, cellular	WiFi, sattelites

Radio waves and spectrum

- What dictates choice of spectrum?
 - **Antenna size**: of the order of wavelength, so lower frequencies need costlier antennas
 - **Multiplexing of users**: two users using same frequency bands interfere, so regulations to enable sharing
 - **Propagation characteristics**: lower frequencies propagate better. LF/MF/HF propagate long distances around earth. VHF reflected by ionosphere. Higher frequencies need line of sight (LOS)
- Common bands
 - 800-900, 1800-1900, 2000-2100 MHz – cellular
 - 2.4 GHz ISM band – WiFi (802.11b/g/n)
 - Good balance between antenna size and propagation

Antennas

- Convert between electromagnetic waves on a wire to EM waves in air and vice versa
 - **Isotropic**: equal radiation in all directions, ideal
 - Simple **dipole** antennas: omni-directional in one plane, figure '8' in other two planes
 - **Directional** antennas: focused in only one direction



Isotropic



Omni/Dipole



Yagi/Directional

Communication using radio waves

- Antenna length $\sim$ wavelength/2
- Voice, Video, bit streams lower frequency band
- Antenna size (few hundreds of meters for 1Mhz signal)
- More attenuation

Carrier wave

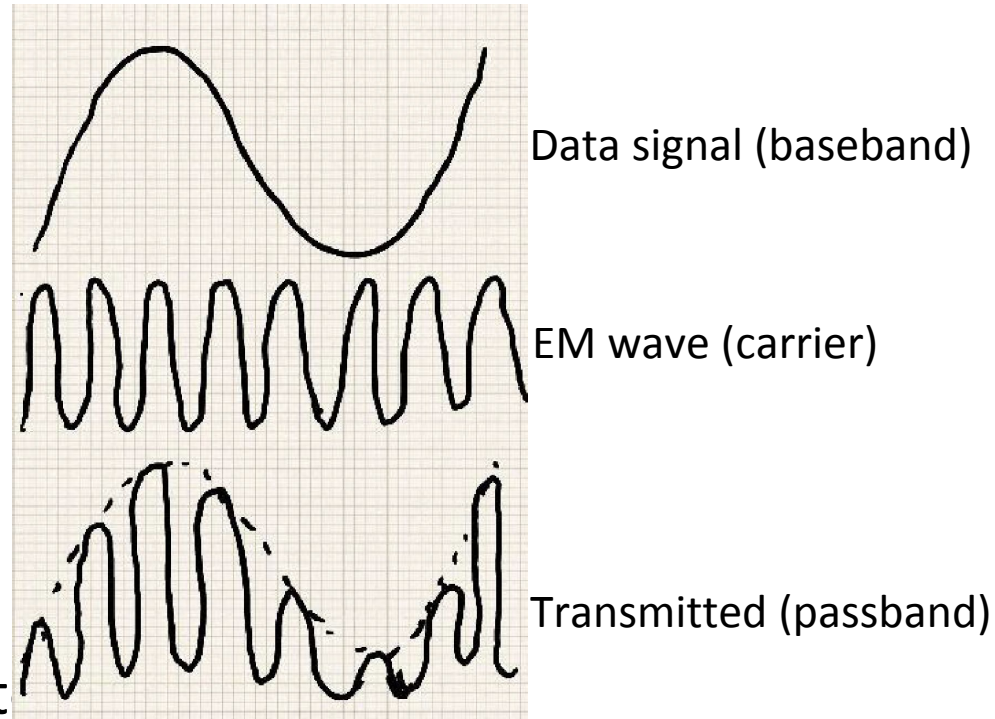
- **Carrier wave** $s = A \cos (2 \pi f t + \phi)$

A = Amplitude, f=Frequency, θ = Phase Period $T = 1/f$, Frequency is measured in Cycles/sec or Hertz

- Carrier signal to carry to long distance
- **Modulation**: selected carrier frequency signal to mix baseband
- Carrier high frequency, smaller antenna size
- Less attenuation

Communication using radio waves

- Transmit information using a sin/cosine wave of a given frequency (carrier)
- Simple idea: analog modulation of signal amplitude.



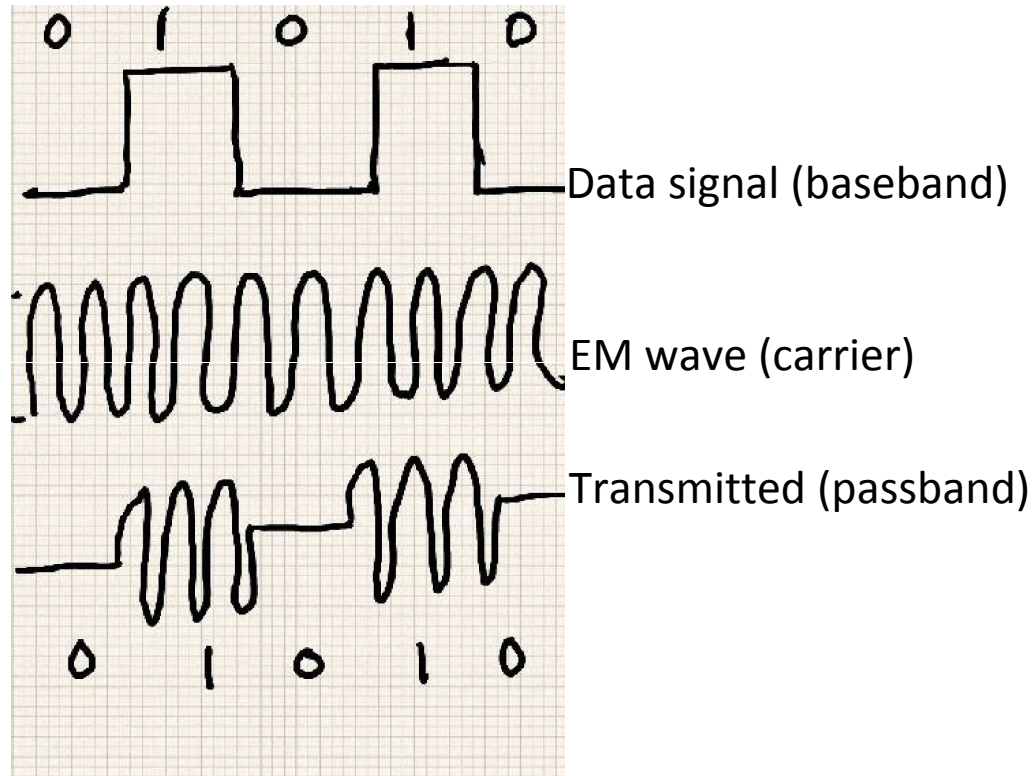
- Used in AM radio
- Error prone and not resilient to

Communication using radio waves (2)

- Better idea – **digital modulation**.
- Convert “signal” into stream of bits.
 - Sample signal at fixed intervals, discrete time signals
 - Quantize the signal value into discrete levels. For example, 4 levels can be represented by 2 bits: 00, 01, 10, 11
- Use bits to modulate the amplitude of the carrier. If 1, send the carrier signal. If 0, no signal. This is amplitude shift keying, there are many more modulation techniques.
- Advantage over analog: even if distorted, can easily tell apart 0 and 1

Communication using radio waves (3)

Amplitude shift keying
(example of digital modulation
Scheme)

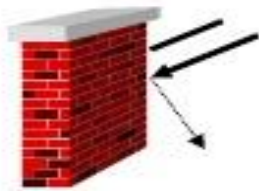


Note: smooth curves for ease, but always think of them as discrete samples.

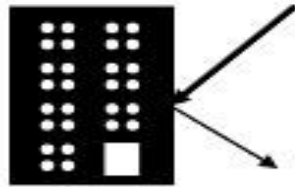
Wireless signal propagation

- What happens when you transmit the modulated signal over the air?
 - Signal travels over multiple paths – multiple copies
 - Each copy of the signal suffers different attenuation
 - Path loss (inverse square law)
 - Reflection, refraction, diffraction, shadowing, scattering etc
 - Each copy of the signal may also have its frequency shifted slightly (Doppler shift due to movement)
 - Finally, there is always background thermal noise
 - Various channel models exist to characterize these effects
- Received signal is sum of multiple different copies of the signal + noise

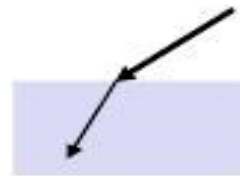
Path Loss Effects



shadowing



reflection



refraction



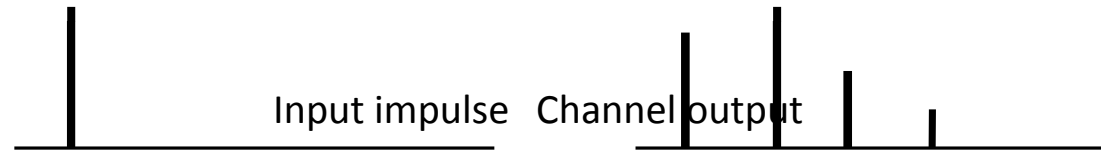
scattering



diffraction

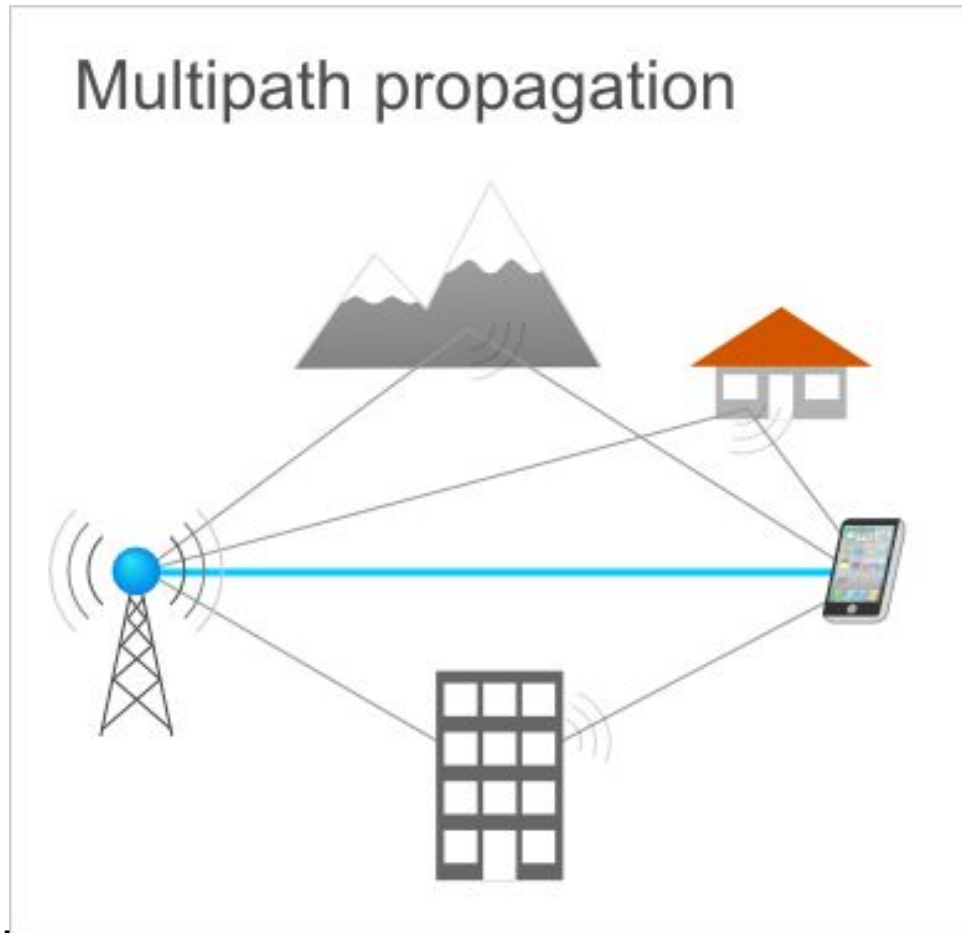
Wireless signal propagation (2)

- Wireless channel is described by **channel impulse response** “h” – what do you receive when you send one impulse.
- Ideally, you only receive the impulse (with some propagation delay).
- With multipath, you receive multiple copies of the impulse, each scaled by different amounts.

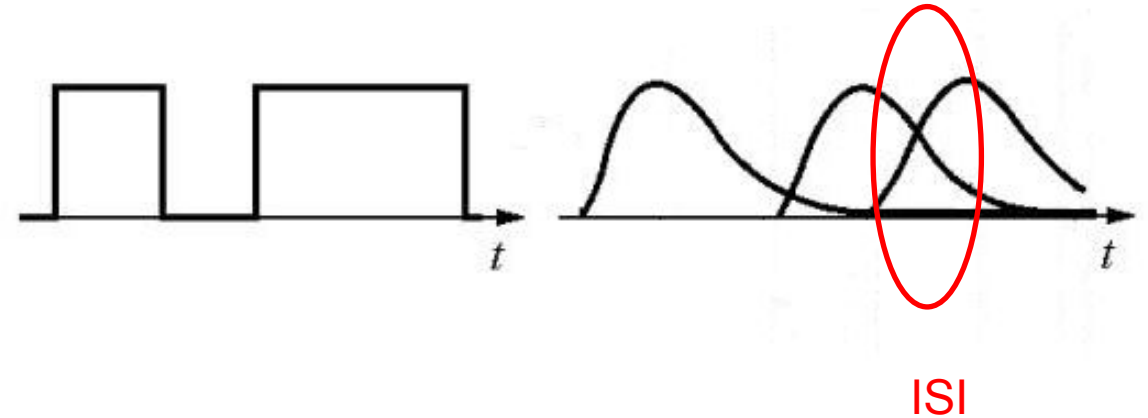


- Transmitted signal “x”, channel “h”, noise “n”, then received signal is $y = h * x + n$
- Here “*” is the convolution operator. That is, for each sample in signal, we consider the impulse response, and add up all these components over all samples.
- Need to estimate and compensate for this “h” at receiver – channel equalization.

Wireless signal propagation (3)



Img src:
http://wiki.yatebts.com/index.php/Radio_Propagation_Concepts



Img src:
<http://durofy.com/pulse-code-modulation-intersymbol-interference-eye-patterns/>

Wireless signal propagation (4)

- **Delay spread**: signals travelling along different paths with diff length will arrive at the receiver at diff times...

Wireless signal propagation (5)

- If each symbol duration is longer than multipath delay spread of channel, all copies of a symbol arrive in the same symbol duration. Easier to estimate the channel “h” and compensate for it at receiver
- If symbol duration < delay spread, copies of previous symbol interfere with current symbol – [inter-symbol interference \(ISI\)](#)
- When large delay spread causes ISI, we need complicated equalization techniques to cancel out effects of all previous symbols (“multi-tap equalization”) at receiver
- As the carrier frequency of a signal is varied, the magnitude of the change in amplitude will vary. The coherence bandwidth measures the separation in frequency after which two signals will experience uncorrelated fading.

Coherence Bandwidth

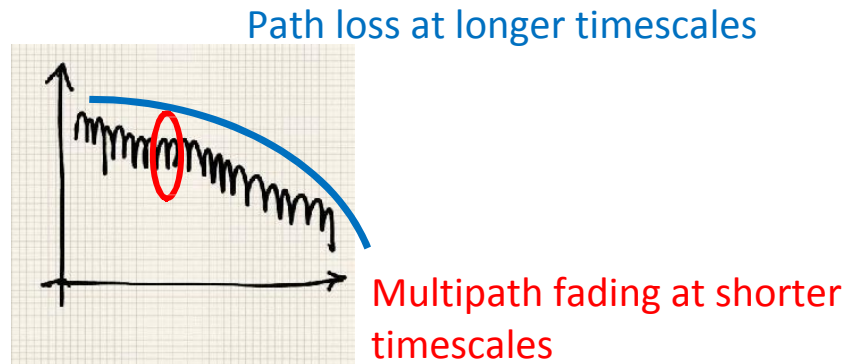
- Coherence bandwidth of a channel – the bandwidth over which channel frequency response H stays the same.
- Coherence bandwidth $\sim 1/\text{DelaySpread}$
- In channels with lot of multipath (large delay spread), coherence bandwidth can be lower than width of channel. Such channels are called [frequency selective channels](#).
- For WiFi, channel bandwidth is 20 MHz. Coherence bandwidth is sometimes lower than this.

Wireless signal propagation (6)

- In **flat fading**, the coherence bandwidth of the channel is larger than the bandwidth of the signal. Therefore, all frequency components of the signal will experience the same magnitude of fading.
- In **frequency-selective fading**, the coherence bandwidth of the channel is smaller than the bandwidth of the signal. Different frequency components of the signal therefore experience uncorrelated fading.

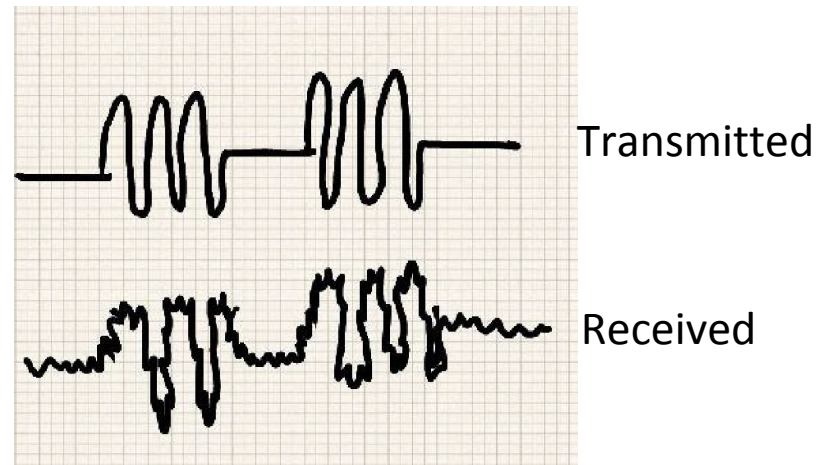
Wireless signal propagation (7)

- Net effect of the wireless channel
 - Signal is attenuated (weakened) at longer timescales due to path loss etc
 - Variations at smaller timescales due to multipath fading (multiple copies combine constructively or destructively)
 - Finally, background thermal noise



Decoding wireless signals

- Let us revisit our example of digitally modulated signal
- The signal in the “1” bits is distorted due to the channel
- The “0” bits aren’t exactly zero either due to noise
- Net effect: 0 and 1 are hard to distinguish. Possible errors in communication.
- If received signal $\gg$ noise, can decode
- If received signal power and noise power are roughly equal, hard to recover the transmitted signal

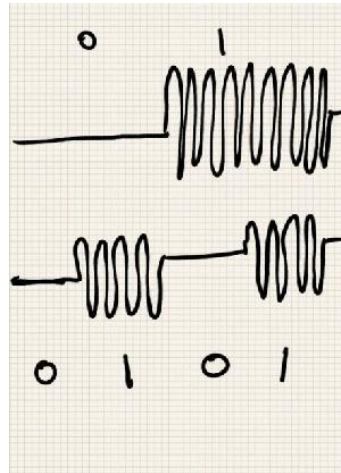


The concept of SNR

- Signal to noise ratio (**SNR**) = Power of signal / power of noise
- SNR is measured in logarithmic **decibel (dB)** units.
- $\text{SNR(dB)} = 10 * \log_{10} (P_s / P_n)$
- For example, $P_s / P_n = 1000$, $\text{SNR} = 30 \text{ dB}$
- Communication systems have a **minimum SNR threshold below which they cannot decode correctly**
- Path loss and any attenuation can also be measured in dB. E.g., signal transmitted at 50 dB, path loss 30 dB, received $50 - 30 = 20 \text{ dB}$ (logarithmic unit, hence subtraction)
- Another unit to measure signal strength: dBm (relative to $P_n = 1\text{mW}$). E.g., if signal transmitted at 1000mW, then signal strength = $10 * \log_{10}(1000/1) = 30 \text{ dBm}$.

How fast can you send?

- SNR decides how easy to decode, and how many errors
- The “symbol duration” (how long you send each 0 and 1) decides the **rate** at which you can send data bits



Slower rate (longer symbols)

Faster rate (shorter
symbols)

- How to decide what rate to send at?
- Math gets harder and harder...

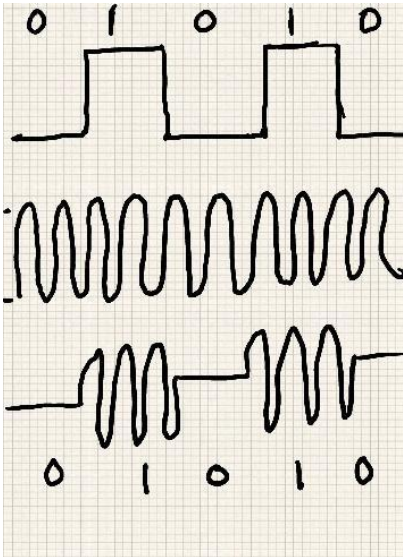
Now, the math: Discrete Fourier Transform

- Fourier transforms are a way to represent and deal with wireless signals. Makes math easier and more intuitive.
- Fourier analysis – a way of representing a signal as a sum of sine and cosine curves of various frequencies (approx.)
- Discrete Fourier Transform (DFT): Fourier transform for discrete time signals (which is what we deal with)
- Given N samples of a time series signal, $x(n)$, $n=0, \dots, N-1$, DFT computes N coefficients, $X(k)$, $k=0, \dots, N-1$, such that, each $X(k)$ represents the weight of a sin/cosine curve of a particular frequency $2\pi k/N$.
- That is, take N sin/cosine curves of N different frequencies ($2\pi k/N$, $k=0, \dots, N-1$), scale them by the DFT coefficients $X(k)$ and add them up, you will get the original time series signal $x(n)$, $n=0, \dots, N-1$.
- Note that I have used very approximate terminology here to avoid equations. For more detail, please refer to any signal processing textbook.

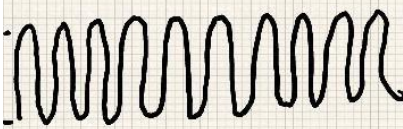
Time domain <-> frequency domain

- The time series of a signal is called the **time domain** representation. The DFT coefficients constitute the **frequency domain** representation. They are both equivalent (if you have one, you can get the other).
- Let's look at our example modulation scheme. The time domain and frequency domain graphs are shown here.
- Note that carrier has only one component (at its frequency) in DFT
- The transmitted signal is "shifted" to the carrier's frequency.

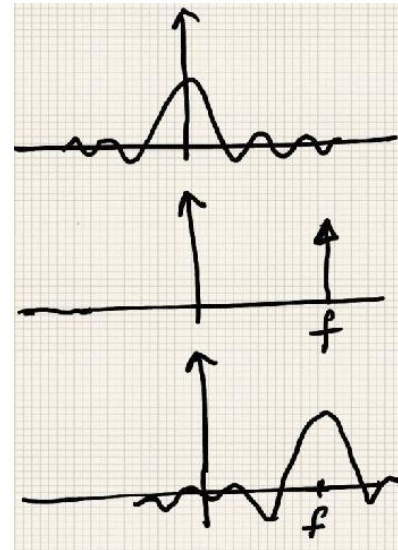
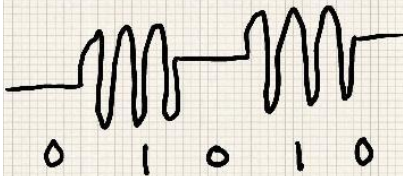
Data signal (baseband)



EM wave (carrier)

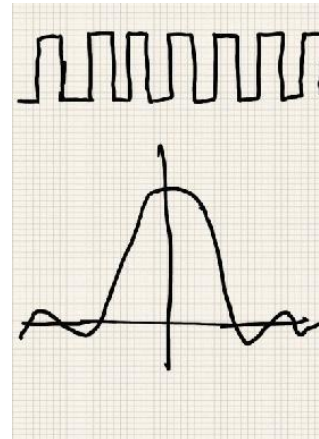
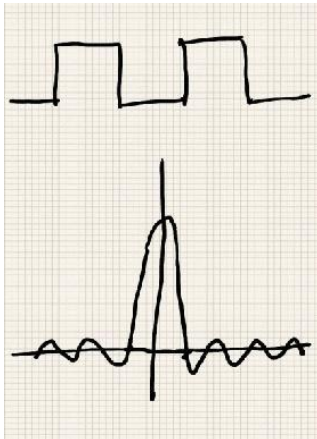


Transmitted (passband)



So, how fast can you send?

- Let's revisit question of how fast you can send. The analysis becomes very simple in frequency domain
- What is the difference between a faster pulse and a slower pulse?
- DFT of faster pulse has a wider frequency "band", i.e., greater **bandwidth** than slower pulse
- Bandwidth is inversely proportional to symbol duration
- More bandwidth -> more frequencies of EM spectrum used



So, how fast can you send? (2)

- How fast you can send depends on two factors
 - Your allocated frequency band. For example, 802.11g has 20 MHz channels allocated. So if you send faster than a certain rate, your frequency domain signal will overlap with your neighbor's signal and lead to data loss. (Note how such an analysis is easy with DFT!)
 - Your hardware capability to sample. More frequencies mean that you must send out more samples. Nyquist's sampling law says that if your highest frequency in baseband is f , you must sample at the rate of at least $2f$. Faster sampling requires faster hardware.
- Once you decide the bandwidth of your signal (based on above two factors), you know how fast your pulses can be

So, how fast can you send? (3)

- **Max data rate**

Bandwidth available

of levels in digital signal

Quality of signal – level of noise

- **Nyquist Theorem** – defines theoretical max bit rate in noiseless channel

Even perfect (noiseless) channels have limited capacity

- **Shannon Theorem** – Nyquist Theorem extended – defines theoretical max bit rate in noisy channel

If random noise is present, situation deteriorates rapidly!

Nyquist law

- Max rate at which digital signal can be transmitted over a channel bandwidth B [Hz] is

$$C_{noiseless} = 2 \times B \times \log_2 M$$

- M – number of discrete levels in digital signal
- $M \uparrow \Rightarrow C \uparrow$, however this places increased burden on receiver – instead of distinguishing one of two possible signals, now it must distinguish between M possible signals

Shannon law

- What is the “speed” of the wireless channel? For example, you have a 100Mbps ethernet link. What is the equivalent for a wireless channel?
- The rate at which you can send information depends on two factors
 - The rate at which you can send symbols, which depends on **bandwidth**
 - How effectively you can distinguish the 0s and 1s in each symbol, which depends on **SNR**

Shannon Law(2)

- Shannon's capacity

$$C = B * \log_2(1 + P_S/P_N)$$

B = bandwidth, P_S/P_N is SNR as a ratio (not dB)

- For example, for a 20 MHz channel, and SNR 30 dB, what is C?
 - SNR = 30 dB, so $P_S/P_N = 1000$
 - $C \sim 20M * \log_2(1001) \sim 20M * 10 \sim 200$ Mbps
- Shannon's capacity is an upper bound. That is, if you try to send at a rate $R > C$, you will see lots of errors.
- That said, any scheme that sends at a rate $R < C$ is not bound to succeed, you must build your signal wisely.
- Next lecture, we will see ways of constructing a transmission signal (modulation) that will get us close to channel capacity.

Summary

- Wireless physical layer uses electromagnetic waves of a certain frequency spectrum
- The bits in the data signal are used to modulate the carrier to get the transmitted signal
- Wireless channel causes path loss, multipath fading, and other effects that weaken the transmitted signal
- Signal to noise ratio (SNR) determines how easy it is to recover the bits from the received signal
- Bandwidth of the transmission determines how fast you can send bits one after the other
- Shannon gives us a way to calculate the maximum rate at which we can send bits (capacity) as a function of bandwidth and SNR